Model for seasonal variation

Parametrization

A model for seasonal variation with periodicity m for the random vector $\S = (x_1, \ldots, x_n)$, $n > m$ is obtained assuming that the sums $x_i + x_{i+1} + \ldots + x_{i+m-1}$ are independent Gaussian with precision τ.

The density for $\mathbf{x}$ is derived from the $n - m + 1$ increments as

$$\pi(\mathbf{x}|\tau) \propto \tau^{\frac{(n-m+1)}{2}} \exp\left\{-\frac{\tau}{2}\sum(x_i + x_{i+1} + \ldots + x_{i+m-1})^2\right\} \quad (1)$$

$$= \tau^{\frac{(n-m+1)}{2}} \exp\left\{-\frac{1}{2}\mathbf{x}^T\mathbf{Q}\mathbf{x}\right\} \quad (2)$$

where $\mathbf{Q} = \tau\mathbf{R}$ and $\mathbf{R}$ is the structure matrix reflecting the neighbourhood structure of the model.

Hyperparameters

The precision parameter τ is represented as

$$\theta = \log\tau$$

and the prior is defined on θ.

Specification

The seasonal model is specified inside the `f()` function as

```
f(<whatever>,model="seasonal",season.length=<season.length>,
              scale.model=<scale.model>, hyper = <hyper>)
```

Hyperparameter spesification and default values

doc `Seasonal model for time series`

hyper

- **theta**
  - **hyperid** `7001`
  - **name** `log precision`
  - **short.name** `prec`
  - **prior** `loggamma`
  - **param** `1 5e-05`
  - **initial** `4`
  - **fixed** `FALSE`
  - **to.theta** `function(x) log(x)`
  - **from.theta** `function(x) exp(x)`

constr `FALSE`

nrow.ncol `FALSE`

augmented `FALSE`

aug.factor `1`

aug.constr

n.div.by

n.required `FALSE`

set.default.values `FALSE`

missing.values.warning `TRUE`

pdf `seasonal`

Example

```
n=203
n.seas=12
trend=1:n
seasonal.sim=rep(1:n.seas, ceiling(n/n.seas))[1:n]

a=1
b=0.5
y = rnorm(n,a+b*trend,1)+rnorm(n,0.2*seasonal.sim,1)

data=data.frame(y=y,trend=trend,seasonal=trend)
formula = y~f(trend,model="rw2")+f(seasonal,model="seasonal",
              season.length=n.seas)
result=inla(formula,family="gaussian",data=data)
```

Notes

The seasonal is intrinsic with rank deficiency $m - 1$.